A MORPHISM OF ALGEBRAIC STACKS ETALE OVER A COMMON BASE IS ETALE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The statement that a morphism of algebraic stacks between separated or DM stacks over a base is DM. Invoked at the start of the counted proof (source lines 8205-8206). Makes the morphism DM so that it admits etale presentations by algebraic spaces.
- The existence of etale presentations for DM morphisms of algebraic stacks. Invoked twice inside the counted proof (source lines 8209-8215). Produces the algebraic spaces V and U presenting the two stacks over the base and over the target.
- The version for algebraic spaces of the etale permanence statement. Invoked at the end of the counted proof (source lines 8225-8228) as a cross-chapter pointer. Gives etaleness of the induced morphism of presenting spaces.

Source reference. Original lemma lemma-DM: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 7800–7800. Navigation pointer only: the counted text refers to the Stacks item lemma-DM, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-compose-after-separated: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1089–1089. Navigation pointer only: the counted text refers to the Stacks item lemma-compose-after-separated, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference spaces-morphisms-lemma-etale-permanence: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4653–4653. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-etale-permanence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{X}, \mathcal{Y}$ be algebraic stacks étale over an algebraic stack $\mathcal{Z}$. Any morphism $\mathcal{X} \to \mathcal{Y}$ over $\mathcal{Z}$ is étale.*

Proof. The morphism $\mathcal{X} \to \mathcal{Y}$ is DM by Lemma lemma-compose-after-separated. Let $W \to \mathcal{Z}$ be a surjective smooth morphism whose source is an algebraic space. Let $V \to \mathcal{Y} \times_{\mathcal{Z}} W$ be a surjective étale morphism whose source is an algebraic space (Lemma lemma-DM). Let $U \to \mathcal{X} \times_{\mathcal{Y}} V$ be a surjective étale morphism whose source is an algebraic space (Lemma lemma-DM). Then

$$U \longrightarrow \mathcal{X} \times_{\mathcal{Z}} W$$

is surjective étale as the composition of $U \to \mathcal{X} \times_{\mathcal{Y}} V$ and the base change of $V \to \mathcal{Y} \times_{\mathcal{Z}} W$ by $\mathcal{X} \times_{\mathcal{Z}} W \to \mathcal{Y} \times_{\mathcal{Z}} W$. Hence it suffices to show that $U \to W$ is étale. Since $U \to W$ and $V \to W$ are étale because $\mathcal{X} \to \mathcal{Z}$ and $\mathcal{Y} \to \mathcal{Z}$ are étale, this follows from the version of the lemma for algebraic spaces, namely Morphisms of Spaces, Lemma spaces-morphisms-lemma-etale-permanence. □